

Definitions and Concepts:

1. **Identity Matrix (I_n):** The identity matrix is a square matrix where all diagonal elements are 1, and all off-diagonal elements are 0. For example:

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

2. **Orthogonal Matrix:** A matrix B is orthogonal if:

$$B^T B = I_n,$$

where B^T is the transpose of B , and I_n is the identity matrix.

3. **Symmetric Matrix:** A matrix B is symmetric if:

$$B^T = B.$$

4. **Euclidean Norm:** The Euclidean norm of a vector v is defined as:

$$\|v\| = \sqrt{v^T v}.$$

If $\|v\| = 1$, the vector is called a **unit vector**.

5. **Rank of a Matrix:** The rank of a matrix is the maximum number of linearly independent rows or columns.

(a) Prove or Disprove Whether $B = I_n - 2(vv^T)$ is Orthogonal and Symmetric

Step 1: Symmetry of B

- Start with the definition of B :

$$B = I_n - 2(vv^T).$$

- To check if B is symmetric, compute B^T (the transpose of B):

$$B^T = (I_n - 2vv^T)^T.$$

- Use the property of transposes:
 - The transpose of I_n is itself ($I_n^T = I_n$).
 - The transpose of vv^T is also itself ($(vv^T)^T = vv^T$) because $v^T v$ is symmetric.

Thus:

$$B^T = I_n - 2vv^T = B.$$

Conclusion: B is symmetric.

Step 2: Orthogonality of B

- A matrix B is orthogonal if $B^T B = I_n$. Since $B^T = B$ (symmetric), we compute BB :

$$BB = (I_n - 2vv^T)(I_n - 2vv^T).$$

- Expand the product:

$$BB = I_n - 2vv^T - 2vv^T + 4(vv^T)(vv^T).$$

- Simplify:
 - $vv^T vv^T = v(v^T v)v^T = (v^T v)vv^T$, where $v^T v$ is a scalar (the norm squared of v).
 - Let $v^T v = \|v\|^2$. Then:

$$BB = I_n - 4vv^T + 4\|v\|^2 vv^T.$$

- For general v , $\|v\|^2$ may not equal 1, so the matrix B may not satisfy $B^T B = I_n$ for arbitrary v .

Conclusion: B is not orthogonal for arbitrary v , but it is symmetric.

(b) Prove or Disprove Whether $C = I_n - 2(ww^T)$ is Orthogonal

Here, it is given that $\|w\| = 1$, so $w^T w = 1$.

Step 1: Compute $C^T C$

- Start with $C = I_n - 2ww^T$.
- Since C is symmetric ($C^T = C$), compute CC :

$$CC = (I_n - 2ww^T)(I_n - 2ww^T).$$

- Expand the product:

$$CC = I_n - 2ww^T - 2ww^T + 4(ww^T)(ww^T).$$

- Simplify:
 - $(ww^T)(ww^T) = w(w^T w)w^T = w(1)w^T = ww^T$ (since $\|w\| = 1$).
 - Substitute back:

$$CC = I_n - 4ww^T + 4ww^T = I_n.$$

Conclusion: C is orthogonal when $\|w\| = 1$.

(c) Show That the Rank of $I_n - vv^T$ is Less Than n When $\|v\| = 1$

Step 1: Analyze the Matrix $I_n - vv^T$

1. Start with:

$$I_n - vv^T.$$

2. Since $\|v\| = 1$, vv^T is a rank-1 matrix because it is formed by the outer product of v with itself, and v is a single vector.

Step 2: Rank of $I_n - vv^T$

- The identity matrix I_n has rank n .
- The matrix vv^T has rank 1 (as explained above).
- Subtracting vv^T from I_n reduces the rank by 1 because the null space of vv^T is of dimension $n - 1$ (all vectors orthogonal to v).

Thus:

$$\text{Rank}(I_n - vv^T) = n - 1.$$

Final Answers

- (a):
 - B is symmetric for any v .
 - B is not orthogonal for arbitrary v .
- (b):
 - C is orthogonal when $\|w\| = 1$.
- (c):

- The rank of $I_n - vv^T$ is $n - 1$, which is less than n , when $\|v\| = 1$.